

So weight of P is greater to get the same tension as Q.

9

D

Let T be the tension in the string.

For vertical equilibrium of mass,

$$T \cos 30.0^\circ = mg \quad (1)$$

For horizontal circular motion of mass,

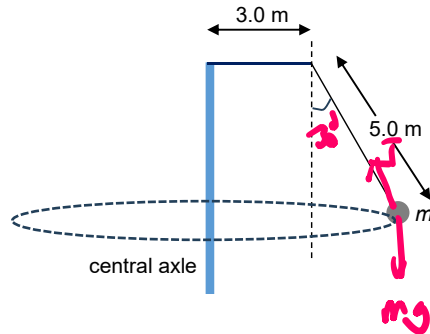
$$T \sin 30.0^\circ = mr\omega^2 \quad (2)$$

$$\frac{(2)}{(1)} \rightarrow \tan 30.0^\circ = r\omega^2/g$$

$$\tan 30.0^\circ = (3.0 + 5.0 \sin 30.0^\circ) \omega^2 / 9.81$$

$$\omega = 1.01 \text{ rad s}^{-1}$$

$$\text{Time for one revolution } T = \frac{2\pi}{\omega} = 6.2 \text{ s.}$$



10

C

Apply Newton 2nd law: